

The purpose of this exercise is to get an understanding of how you code and provide you with a chance to experience the type of work you will be doing at Branch. We do not expect this assessment to take any longer than 3-5 hours; if it takes much longer please stop and send what you have completed.

Platform Engineer

A recently signed customer wants to integrate a subset of GitHub's data into their application. We have discussed their needs and they want an endpoint they can provide a username that will then return the data in JSON format as specified below (that also serves as an example):

```
{
  user_name: "octocat",
  display_name: "The Octocat",
  avatar: "https://avatars.githubusercontent.com/u/583231?v=4",
  geo_location: "San Francisco",
  email: null,
  url: "https://api.github.com/users/octocat",
  created_at: "Tue, 25 Jan 2011 18:44:36 GMT",
  repos: [{
    name: "boysenberry-repo-1",
    url: "https://api.github.com/repos/octocat/boysenberry-repo-1"
  }, ...
]
```

Getting Started: <https://docs.github.com/en/rest/guides/getting-started-with-the-rest-api>

Data Sources: <https://api.github.com/users/octocat> | <https://api.github.com/users/octocat/repos>

The example response above is the result of calling the API with the username "octocat". The data is merged after calling the two APIs noted. Be sure to take note of the difference(s) in parameter names as well as any potential formatting differences between GitHub's APIs and the expected response.

No token or signup is necessary to use these Github APIs; however, you can be rate limited. Perhaps implementing a caching mechanism might help? Of course, you could get an access token that could be set at runtime (we do not expect this).

In Summary

- Stand up a server
- Have an endpoint that takes a **username**
- Fetch or retrieve the data
- Return the JSON defined above
- Have tests to prove your implementation

Push your finalized code to a public repo (GitHub, BitBucket, GitLab). Provide a `README` explaining your decisions, architecture, and how to install/run and utilize your service.

We look forward to seeing your code!

What language/tools do I use? Can I use a framework or library?

Our core services are written in Java/Spring Boot; therefore, we ask you to complete the assignment utilizing those two languages. Feel free to use any other libraries that you wish.

What am I being evaluated on exactly?

We want to see production-ready code. Primary criteria are as follows:

- **Organization:** Is the code well organized where it can be easily extended? Is it properly broken apart into logical and maintainable pieces?
- **Readability:** Can the code be easily read and followed by a new developer. Better yet, would *you* be able to even understand it if you come back to it months later?
- **Libraries/Frameworks:** Are these good choices? Is the reasoning sound?
- **Documentation:** Can we run and use the code based on the [README](#)? Is the code itself well documented in some way?
- **Depth of Knowledge:** Are the right data structures being used? Are the strengths of the language chosen being utilized? Does the API follow industry standards?
- **Stability/Security:** Can the code handle bad input? Is the code programmed defensively?

Can I have someone help me?

Please no. We expect the code to reflect your choices and knowledge. That said, feel free to use online resources as needed for reference.